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Malware Analysis Report: Agent Tesla

An edited excerpt from a malware analysis paper written for CSEC 202 (Professor Jonathan S. Weissman, 2024), revised and expanded with original research.

For this report, I chose to analyze a prevalent strain of malware that appears frequently across the MalwareBazaar database. Its wide distribution is likely attributable to its source code having been leaked or publicly released. Agent Tesla is categorized as a Remote Access Trojan (RAT), and given the rapid emergence of new variations, it presented a compelling and timely subject for analysis (abuse.ch).

The specific strain of Agent Tesla which will be analyzed is identified by its SHA-256 hash: 2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2. Links to references will be provided at the end of the report. This analysis was conducted within a Windows 10 Virtual Machine using the following tools: Strings64, UPX, PEiD, PEview, VirusTotal, Regshot, Process Monitor, ApateDNS, Wireshark, x64 Debugger, and IDA Pro.

Static Analysis - Strings

Initial static analysis was performed using the Strings tool to identify readable function names and literal strings embedded in the executable. The output produced approximately one million lines of text, the majority of which were undecipherable, consistent with deliberate obfuscation. After filtering, several thousand lines of meaningful content remained, revealing a broad and invasive function set.

[Figure 1 omitted due to submission size constraints. Screenshot of Strings64 large text output showing the initial results of running the tool against the Agent Tesla executable.]

Among the most notable discoveries were clipboard interaction functions, including GetClipboardData, OpenClipboard, IsClipboardFormatAvailable, CloseClipboard, CountClipboardFormats, EmptyClipboard, and SetClipboardData. The presence of this entire clipboard API cluster indicates the malware is capable of reading, modifying, and wiping clipboard contents which is a dangerous capability since RATs typically operate with keyloggers.

[Figure 2 & 3 omitted due to submission size constraints. Strings output showing the cluster of clipboard-related Windows API function calls referenced in the executable, including GetClipboardData and SetClipboardData.]

Beyond clipboard functions, the strings analysis revealed an extensive list of system-level capabilities such as FILECOPY, FILEDELETE, FILEMOVE, SHELLEXECUTE, SOUNDPLAY, and SHUTDOWN, among thousands of others. The sheer breadth of functions set makes clear that this is not a precision tool but rather designed to grant an attacker comprehensive access to a system.

[Figures 4 & 5 omitted due to submission size constraints. Two side-by-side screenshots of Strings output showing additional pages of system-level function calls, illustrating the extensive capability set embedded in the executable.]

Packing Analysis and PE Structure analysis - UPX, PEiD, PEview

The large amounts of strings warranted an investigation into whether the executable was packed. UPX was run against the sample but failed to unpack it, indicating that it was not packed using UPX format

[Figure 6 omitted due to submission size constraints. Screenshot of UPX output confirms that the tool was unable to unpack the executable, indicating it wasn't compressed using UPX packer.]

PEiD was subsequently used with hardcore scanning enabled. The results confirmed no packer was detected, validating the strings output from the initial analysis was accurate of the actual binary contents.

[Figure 7 omitted due to submission size constraints. PEiD analysis result showing no packer detected, with hardcore scanning mode enabled.]

Loading the executable in PEview confirmed that the portable executable does not embed additional binaries within its resource section. The absence of content in the ".rsrc" section indicates that the malware's functionality is delivered entirely through the single executable.

[Figure 8 omitted due to submission size constraints. PEview screenshot showing the PE file structure, with the .rsrc section visible and empty, confirming no additional binaries are embedded within the executable.]

Threat Intelligence - VirusTotal

VirusTotal provided critical context, though several findings were incomplete at the time of submission, this reflects that this strain is relatively new. The detection tab revealed only 25 of 72 vendors flagged the executable as malicious. Notably absent were Bitdefender, CrowdStrike Falcon, VIPRE, and SentinelOne which underscores the evasion capabilities of newer Agent Tesla variants. The details tab confirmed the SHA-256 hash matched MalwareBazaar, with the first file submitted to VirusTotal only four hours prior. The executable's filename, INV00932024.exe, was noted for future cross-reference.

The relations tab presented associated IPs and FQDNs, most appearing benign on initial review which is consistent with newly established infrastructure. A few domains were later confirmed malicious through dynamic analysis and are cross-referenced in subsequent sections. Of particular note in the Dropped Files section were two files "myriopodous" and "camellin" which became increasingly relevant during dynamic analysis. The sandbox behavior classifications identified the sample as a MALWARE STEALER TROJAN EVADER. The community tab confirmed the broader Agent Tesla family is extensively documented, with over 80,000 related file hashes catalogued, and Discord identified among distribution domains which is consistent with its known use as a webhook-based exfiltration channel.

[Figures 9-19 omitted due to submission size constraints. Screenshots from VirusTotal's detection, details, relations, behavior, and community tabs supports the findings above.]

Threat Intelligence - VirusTotal

Regshot was run twice to confirm consistency, and registry delta results were compared against the host machine as a baseline. A significant finding appeared, the malware accessed and modified a registry entry associated with SMB1 which is a legacy network file-sharing protocol deprecated in modern Windows versions due to well-documented vulnerabilities. The SMB1 usage counter incremented from hex value E1 (225) to E2 (226) immediately after execution and was subsequently observed incrementing again to E3 (227). This indicated a persistent background process repeatedly attempting to access the protocol. Cross-referencing the registry path between the VM and the host confirmed the path was entirely absent on the clean machine, directly correlating its presence to the malware. This suggests a possible lateral movement capability with the SMB1 exploitation.

[Figures 20-22 omitted due to submission size constraints. Regshot output, Registry Editor, and host/VM comparison screenshots supporting the SMB1 findings described above.]

Process Monitoring - Process Monitor

Process Monitor, configured with targeted filters to isolate malware-related activity, captured a large amount operations executed upon running the executable. The volume of operations was substantial, but the analysis ultimately revealed two key output files: "myriopodous" and "camellin," both written to the \AppData\Local\Temp directory. These files are assessed to be the repositories for data collected by the malware this is likely containing harvested credentials, keylog data, or exfiltration staging content. Critically, both filenames appeared in VirusTotal's Dropped Files section during earlier analysis, with "camellin" specifically flagged as malicious. When accessed directly, both files are presented as encrypted and unreadable, confirming the malware's use of encryption to protect exfiltrated data from inspection.

[Figures 23-36 omitted due to submission size constraints. Process Monitor screenshots capturing the operation chain following malware execution, filter configuration, and File Explorer view of the resulting encrypted files "myriopodous" and "camellin" dropped in \AppData\Local\Temp.]

Network Analysis - ApateDNS and Wireshark

ApateDNS revealed that upon execution the malware immediately issued a DNS request to "mail.cmcapama.top" which is a domain previously noted in VirusTotal's Contacted Domains section. A secondary FQDN check returned no results, indicating a single C2 server with no failover architecture. Wireshark findings shared a similar note, capturing the corresponding DNS and ICMP packets confirming the single C2 channel.

[Figures 37-39 omitted due to submission size constraints. ApateDNS and Wireshark screenshots confirming the C2 DNS request to "mail.cmcapama.top" and absence of a secondary FQDN.]

VirusTotal Revisited - C2 Infrastructure Cross-Reference

Returning to VirusTotal to analyze “mail.cmcapama.top” and its associated IP confirmed that all executables connecting to this domain were flagged as malware. While the domain and IP themselves were only flagged by two vendors, consistent with recently established C2 infrastructure. INV 00932024.exe appeared among the connecting executables, directly tying the given name back to the analyzed sample.

[Figures 40-42 omitted due to submission size constraints. VirusTotal analysis of the C2 FQDN, connected executable list, and associated IP address results.]

Disassembly and Debugging - x64 Debugger and IDA Pro

Disassembly and debugging proved to be the most technically challenging phase of the analysis. Upon loading the executable in the x64 Debugger, an unusual error appeared that was never observed during normal execution. This suggested the malware was detecting the debugging environment and altering its behavior accordingly.

[Figure 43 omitted due to submission size constraints. x64 Debugger screenshot showing an error message that appeared only during debugging and not during normal execution.]

All Windows security settings including Virus and Threat Protection, App and Browser Control, and Device Security were disabled in an attempt to isolate the cause. None of those changes altered the malware's behavior during debugging, pointing to a possible anti-analysis mechanism. Attempting to reach specific functions identified during disassembly were also unsuccessful. The CreateFileW function which is logically the first Windows API call a portable executable, would make upon attempting to create or modify files for persistence was not reached during live debugging. IDA Pro confirmed the function's presence within a wrapper function labeled `__createFile`, with a call address of 00437FCB. The address was confirmed to exist in the debugger, but the instructions at that address consisted entirely of anomalous "add byte ptr ds:[eax],al" entries. Which upon further research online, it made clear that is a pattern inconsistent with a legitimate function body and indicative of deliberate obfuscation or self-modification.

[Figures 44-48 omitted due to submission size constraints. IDA Pro and x64 Debugger screenshots showing the CreateFileW function, the __createFile wrapper at address 00437FCB, and the anomalous 'add byte ptr ds:[eax],al' instructions found at that address.]

Subsequent research confirmed two anti-analysis mechanisms documented by Acronis Detection and Response. First, the malware queries registry values to detect whether it is running inside a virtual machine. This behavior consistent with the numerous registry operations mentioned in Process Monitor. Second, the malware uses a crypter called AspireCrypt that disrupts debugging by obfuscating the executable's runtime behavior (Acronis, Inc.). The behavior observed across both tools is consistent with the documented evasion mechanisms mentioned. In addition, the Acronis-documented variant of Agent Tesla also communicated with a mail server for C2, the only difference being in the specific address used. This suggests that the Agent Tesla source code has been made public or is sold as a configurable kit, with individuals substituting their own C2 infrastructure while retaining the core functionality.

Summary

Agent Tesla is an extremely capable Remote Access Trojan whose features include keylogging, clipboard data theft, screenshot capture, webcam access, file system manipulation, and encrypted data exfiltration via a mail-based C2 channel. This analysis identified a newer strain (SHA-256: 2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2) with a low detection rate across major vendors. This is consistent with recently deployed infrastructure and a crypter-obfuscated binary that actively evades both signature-based detection and dynamic analysis.

Key findings include clipboard and system-level API enumeration indicating comprehensive access capabilities. SMB1 was also detected via registry manipulation suggesting potential lateral movement vector. Encrypted file drops ("myriopodous" and "camellin") were confirmed as malicious via cross-reference with VirusTotal. A single C2 FQDN (mail.cmcapama.top) with limited vendor flagging and confirmed anti-VM and anti-debugging mechanisms via AspireCrypt. The behavior of this malware strain is consistent with commercially distributed or publicly leaked RAT kit, with a configurable C2 address.

Removal Recommendations

If Agent Tesla is suspected to be running on a system that antivirus tools have failed to detect, the zero-trust approach is strongly recommended. A clean Windows reinstallation, and credentials stored or entered on the machine (including browser-saved passwords) should be treated as compromised and changed immediately. If full reinstallation is not possible, the following steps may reduce risk:

1. Isolate the infected system immediately by disconnecting from all networks to prevent C2 communication and lateral spread.
2. Run Malwarebytes Anti-Malware in conjunction with Windows Defender for layered detection coverage.
3. Restore from a clean system restore point before the execution of the malware, if available.
4. Apply all pending Windows and antivirus updates to improve detection capability.
5. Change all passwords for system accounts, applications, and websites, and enable two-factor authentication where possible.
6. Monitor the outbound DNS traffic for connections to mail-related FQDNs which may seem suspicious. Persistent connections to unfamiliar mail servers indicate the malware remains active.

References

abuse.ch. "Malware Sample Entry for SHA256: 2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2." MalwareBazaar, 2024, <https://bazaar.abuse.ch/sample/2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2/>. Accessed 30 Apr. 2024.

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